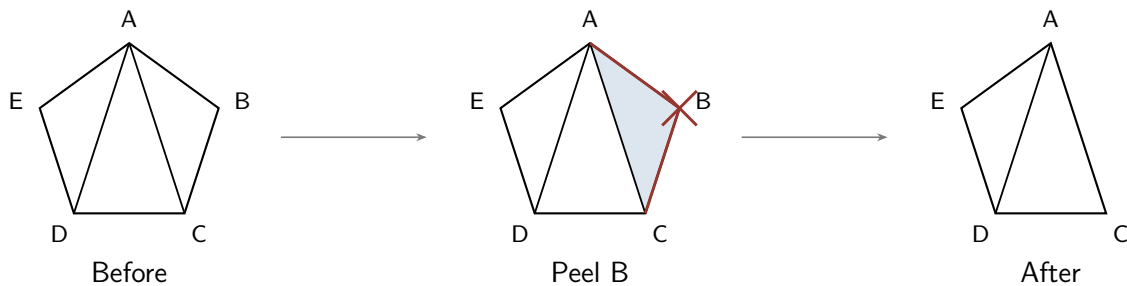
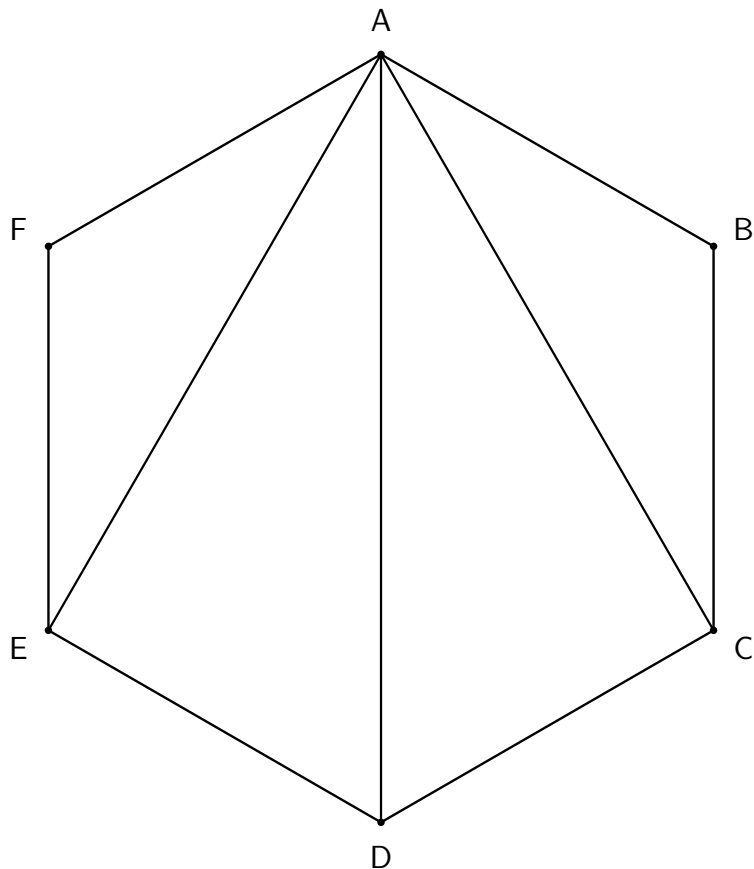


An ear is a triangle with two sides on the outside edge. Peel it by erasing those sides and their meeting corner. Keep its third side. Trace a drawing, then lift the tracing off this page before peeling.



Problem 1: Peel ears from each large drawing on these two pages until one triangle remains, writing the peeled corner letters in order. Can your choices leave different last triangles?



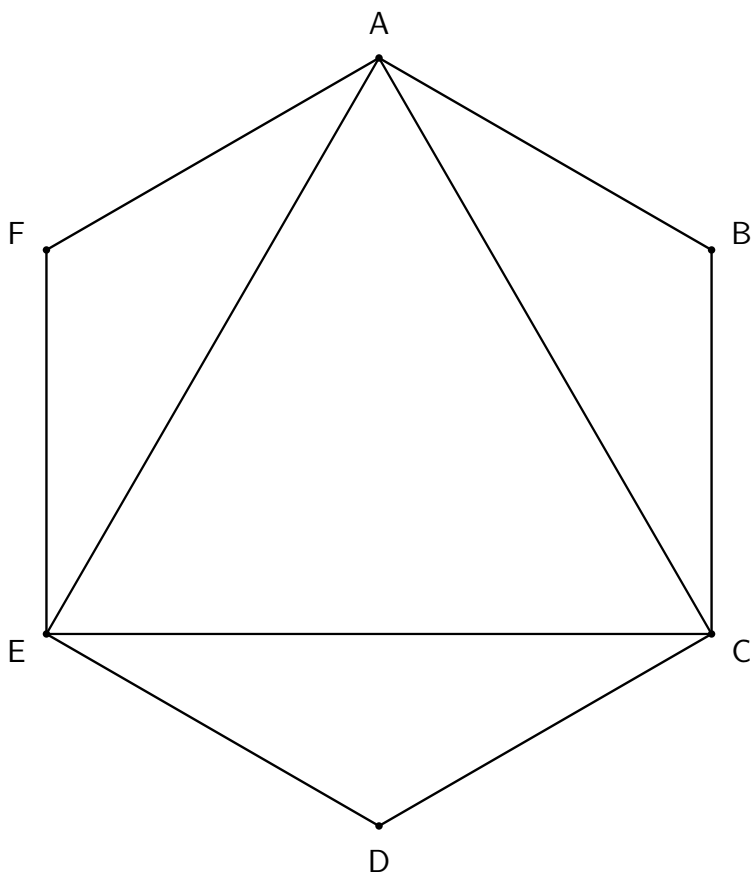
Peeled corners: _____

Last triangle: _____

Peeled corners: _____

Last triangle: _____

Extra workspace for Problem 1



Peeled corners: _____

Last triangle: _____

Peeled corners: _____

Last triangle: _____

Peeled corners: _____

Last triangle: _____

Peeled corners: _____

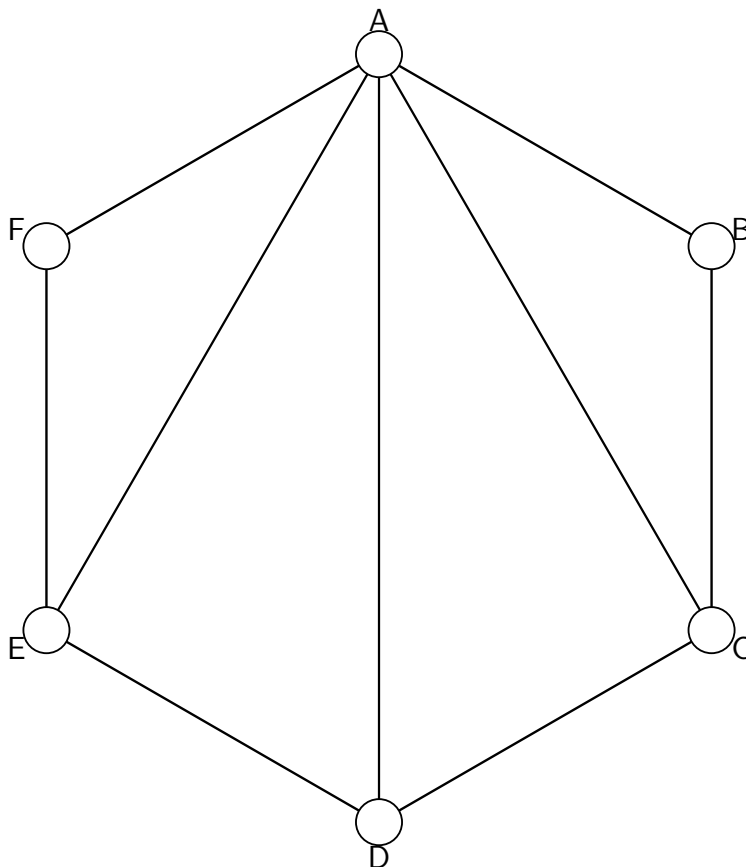
Last triangle: _____

Peeled corners: _____

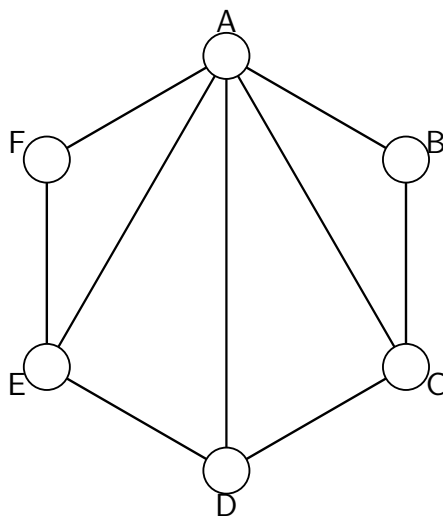
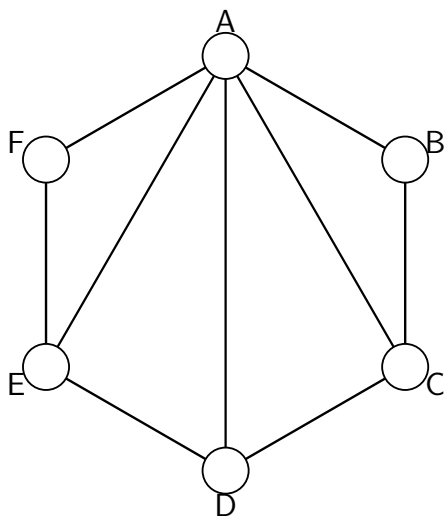
Last triangle: _____

Use red, blue and yellow. Give each corner one color.

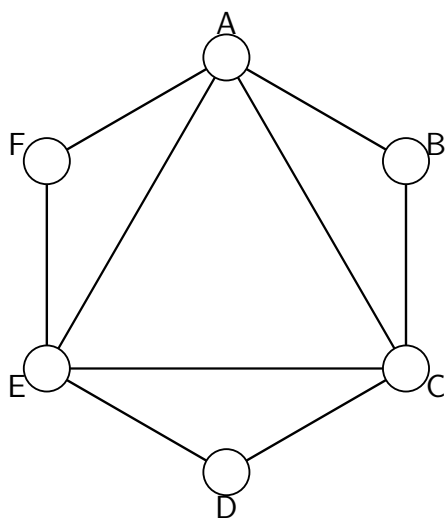
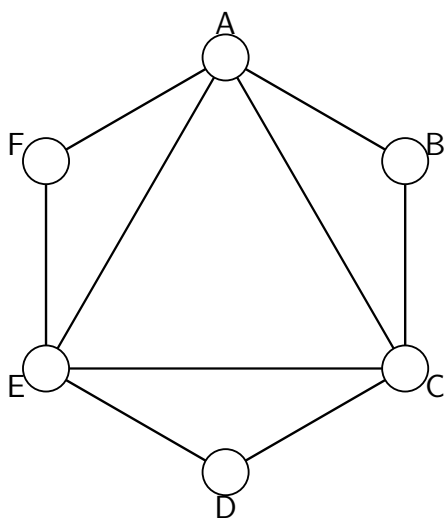
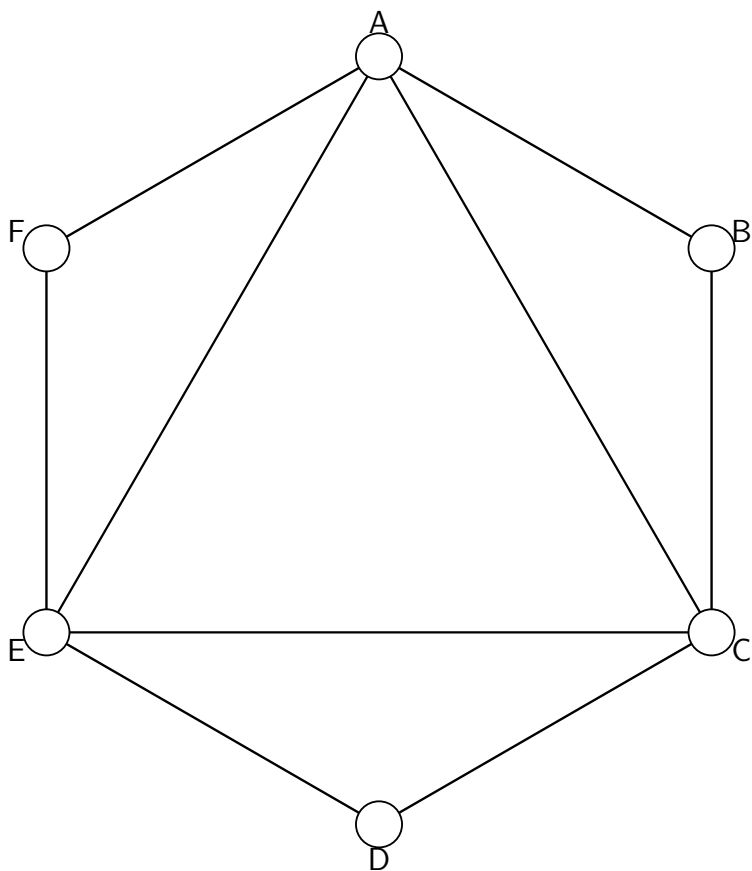
Problem 2: Color the corners of each large drawing on these two pages so every triangle has all three colors. Circle as few corners as possible so every triangle touches a circled corner.



Extra workspace

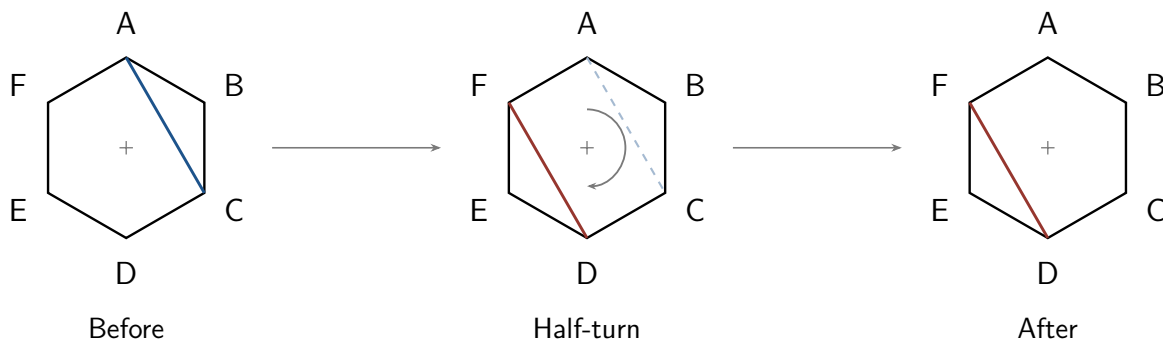


Extra workspace for Problem 2

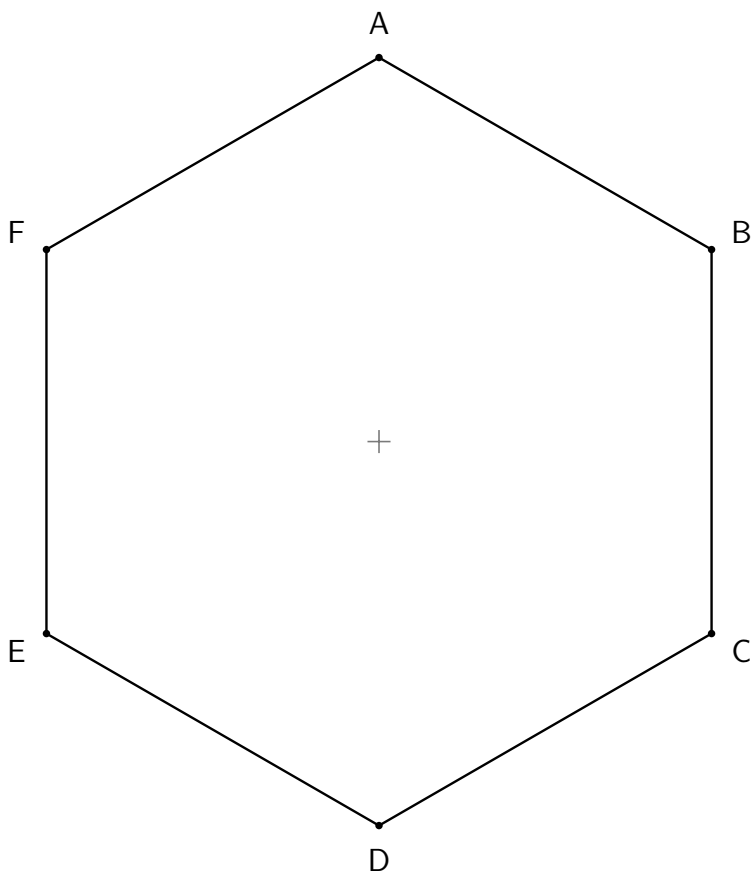


Join corners with straight lines to divide the hexagon into triangles. Lines may meet only at corners. Keep the corner letters fixed; different sets of lines count as different drawings.

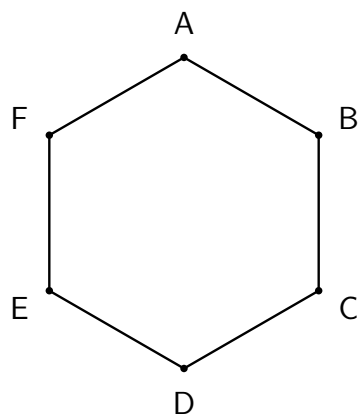
Trace the inside lines and the center mark. Turn the tracing about that mark. It matches when every turned line lands on an original line.



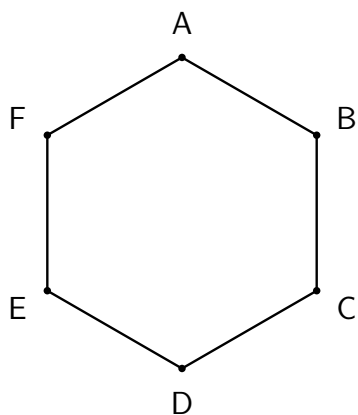
Problem 3: Find all triangulations that match after a half-turn, a one-third-turn or a one-sixth-turn. Save each with its turn on the next page. Explain why each list is complete.



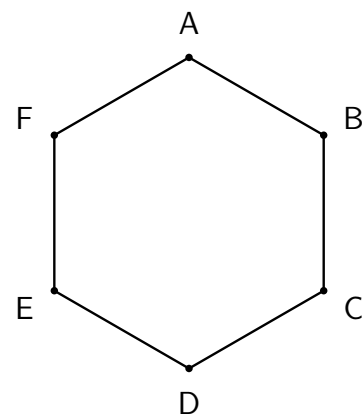
Extra workspace for Problem 3



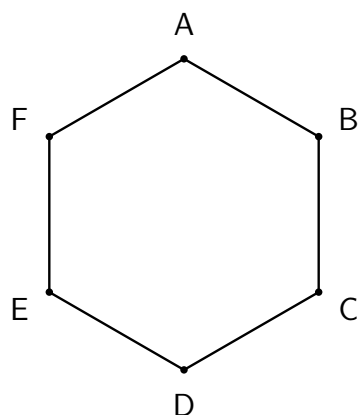
Turn: _____



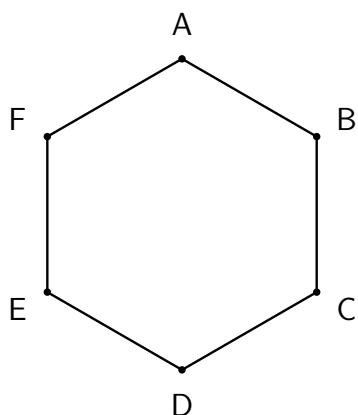
Turn: _____



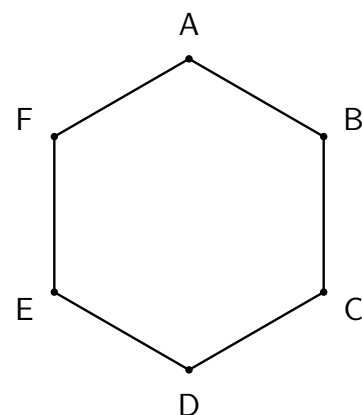
Turn: _____



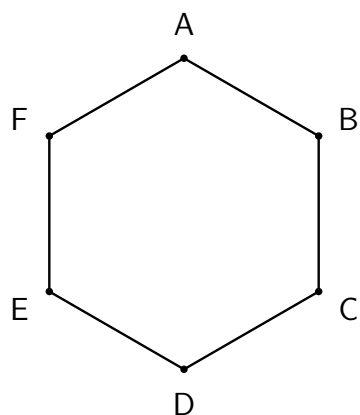
Turn: _____



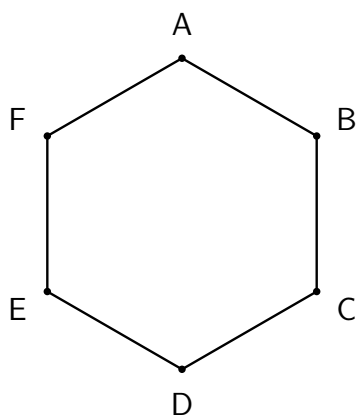
Turn: _____



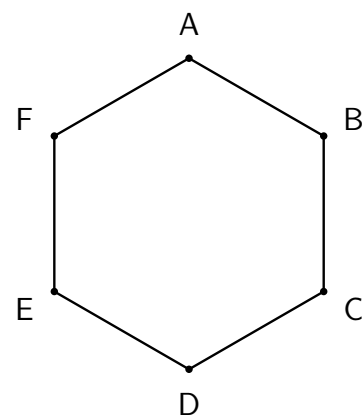
Turn: _____



Turn: _____



Turn: _____



Turn: _____

Explanation:
